Lecture 18: Social media and society

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Sociology 204: Social Networks
Princeton University

Social media:

- Lecture 17: Social media and individuals
- Lecture 18: Social media and society
- Lecture 19: Social media and democracy
- Lecture 20: Fixing social media

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- Allcott et al. do a randomized controlled trial to measure the effect of FB on a sample of American users
- Being off FB for one month leads to 1) reduced online activity and increased offline activity 2) reduced factual news knowledge and political polarization 3) increased subjective well-being and 4) caused a large persistent reduction in post-experiment Facebook use
- Being off FB for one month also leads to new awareness about the good and bad aspects of FB

Now it is your turn: self-experimentation and social media

Self-experimentation and social media

- Assignment 5: Self-experimentation and social media, design

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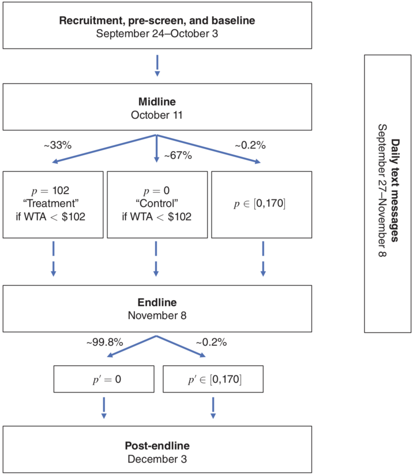
- Assignment 7: Self-experimentation and social media, results
- Assignment 8: Self-experimentation and social media, reflect



Surprises From Self-Experimentation: Sleep, Mood, and Weight

Seth Roberts

A few notes on self-experimentation

- Long history in medicine: `https://en.wikipedia.org/wiki/Self-experimentation_in_medicine`

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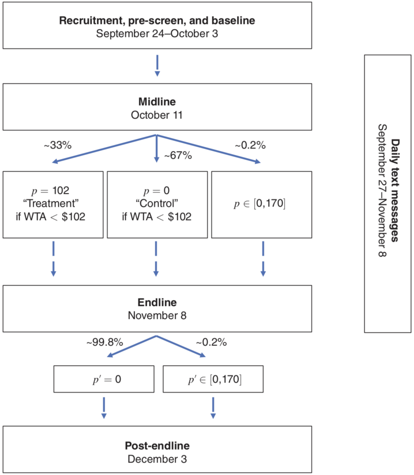
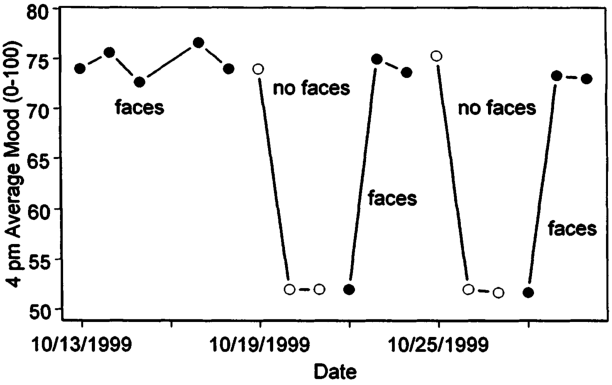
- Not blinded
- Can't learn about heterogeneity across people
- Can learn about unexpected outcomes
- Can iterate quickly





You will be your own control group (“within-subjects design”)

A few notes about this assignment:

- Choose an interesting and important intervention. Best ones will likely be either motivated by science or wellbeing.

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- If this assignment causes problems, come talk to us or University Health Services.

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Background

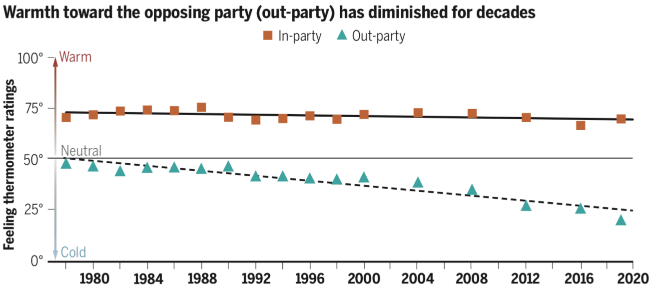
SOCIAL SCIENCE

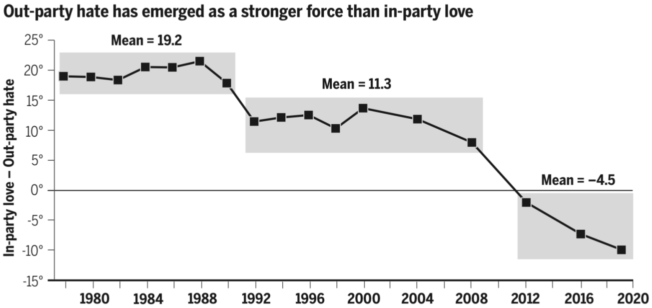
Political sectarianism in America

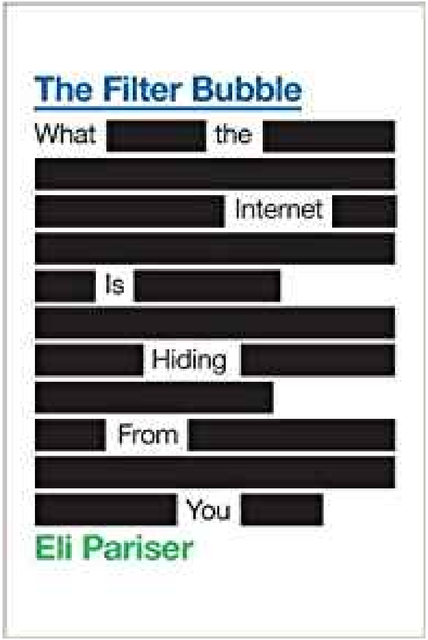
A poisonous cocktail of othering, aversion, and moralization poses a threat to democracy

By **Eli J. Finkel[1], Christopher A. Bail[2], Mina Cikara[3], Peter H. Ditto[4], Shanto Iyengar[5], Samara Klar[6], Lilliana Mason[7], Mary C. McGrath[1], Brendan Nyhan[8], David G. Rand[9], Linda J. Skitka[10], Joshua A. Tucker[11], Jay J. Van Bavel[11], Cynthia S. Wang[1], James N. Druckman[1]**

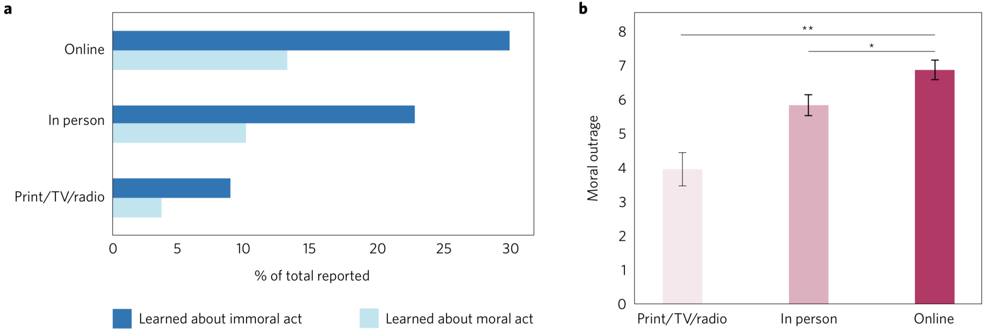
Published in 2020


Warmth toward the opposing party (out-party) has diminished for decades
In-party
Out-party
Feeling thermometer ratings
100°
75°
50°
25°
0°
Warm
Neutral
Cold
1980
1984
1988
1992
1996
2000
2004
2008
2012
2016
2020

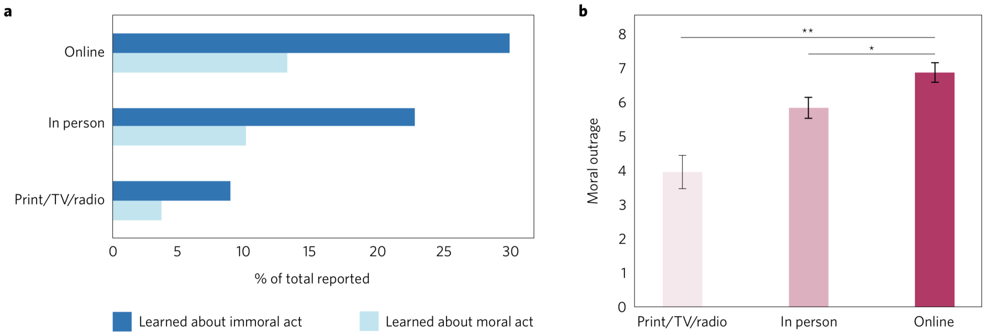

Out-party hate has emerged as a stronger force than in-party love
In-party love – Out-party hate
25°
20°
15°
10°
5°
0°
-5°
-10°
-15°
Mean = 19.2
Mean = 11.3
Mean = –4.5
1980
1984
1988
1992
1996
2000
2004
2008
2012
2016
2020


The Filter Bubble
What the
Internet
Is
Hiding
From
You
Eli Pariser

What goes viral? Moral outrage and lies



- People more likely to learn about immoral acts online than through other media



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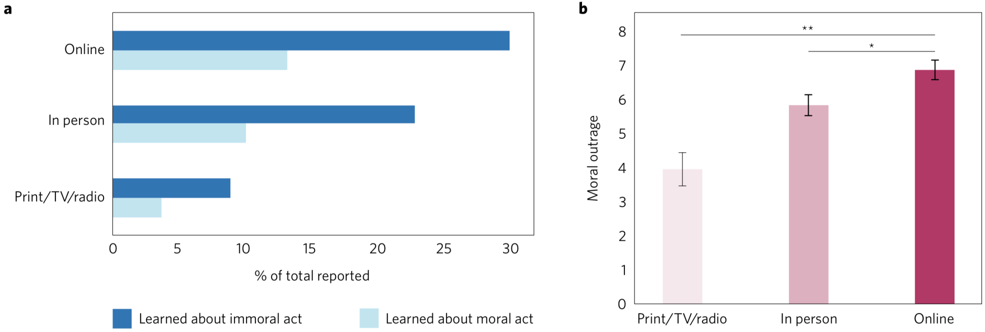
- Imoral acts that people see online lead to more outrage



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- Imoral acts that people see online lead to more outrage
- "Digital media transform moral outrage by changing both the nature and prevalence of the stimuli that trigger it."

SOCIAL SCIENCES

How social learning amplifies moral outrage expression in online social networks

William J. Brady[1]*, Killian McLoughlin[1], Tuan N. Doan[2], Molly J. Crockett[1]*

- two observational studies and two experiments (all preregistered)

SOCIAL SCIENCES

How social learning amplifies moral outrage expression in online social networks

William J. Brady[1]*, Killian McLoughlin[1], Tuan N. Doan[2], Molly J. Crockett[1]*

- two observational studies and two experiments (all preregistered)
- reinforcement learning and norm learning, both of which are controlled in part by design of social media platforms

“A person can be viewed as expressing moral outrage if:

- they have feelings in response to a perceived violated of their personal morals
- their feelings are comprised of emotions such as anger, disgust, and contempt
- the feelings are associated with specific reactions including blaming people/events/things, holding them responsible or wanting to punish them.”

They labeled a many example tweets and then built a Digital Outrage Classifier

https://www.youtube.com/watch?v=b2AYlD8ReeA&t=11m26s

Supporting our hypotheses, we found that daily outrage expression was significantly and positively associated with the amount of social feedback received for the previous day's outrage expression (Study 1: $b = 0.03$, $p < .001$, 95% CI = [0.03, 0.03]; Study 2: $b = 0.02$, $p < .001$, 95% CI = [0.02, 0.03]). For our model, this effect size translates to an expected 2-3% increase in outrage expression on the following day of tweeting if a user received a 100% increase in feedback for expressing outrage on a given day. For instance, a user who averaged 5 likes/shares per tweet, and then received 10 likes/shares when they expressed outrage, would be expected to increase their outrage expression on the next day by 2-3%. While this effect size is small, it can easily scale on social media over time, become notable at scale at the network level, or for users who maintain a larger followership and could experience much higher than 100% increases in social feedback for tweeting outrage content (e.g., political leaders). For other model specifications to test the robustness of the effect, see SOM, Section 2.0.

Three main findings from observational studies

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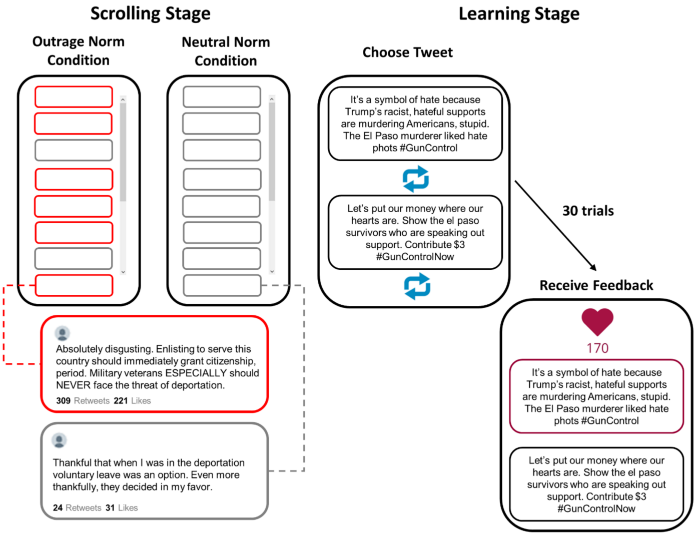
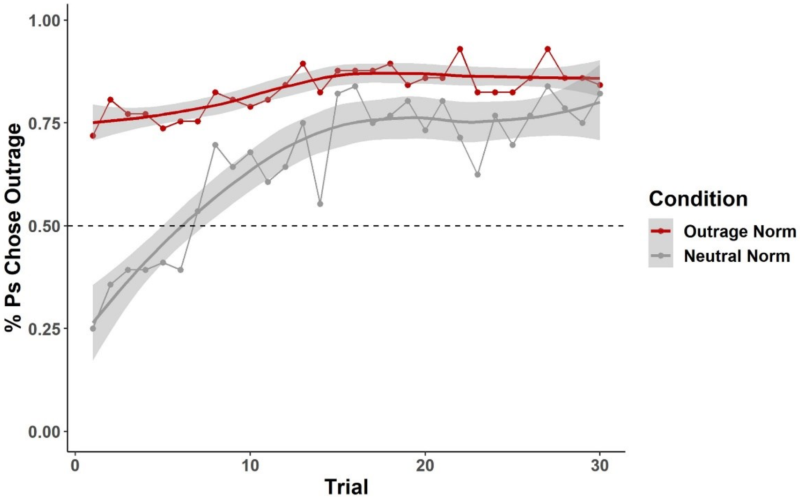
- “users are more likely to express outrage in more ideologically extreme social networks”
- “in more ideologically extreme social networks, users’ outrage expression behavior is less sensitive to social feedback.”

Finding are consistent with reinforcement learning and norm learning, but there are limits to what they can learn just from watching without controlling the environment



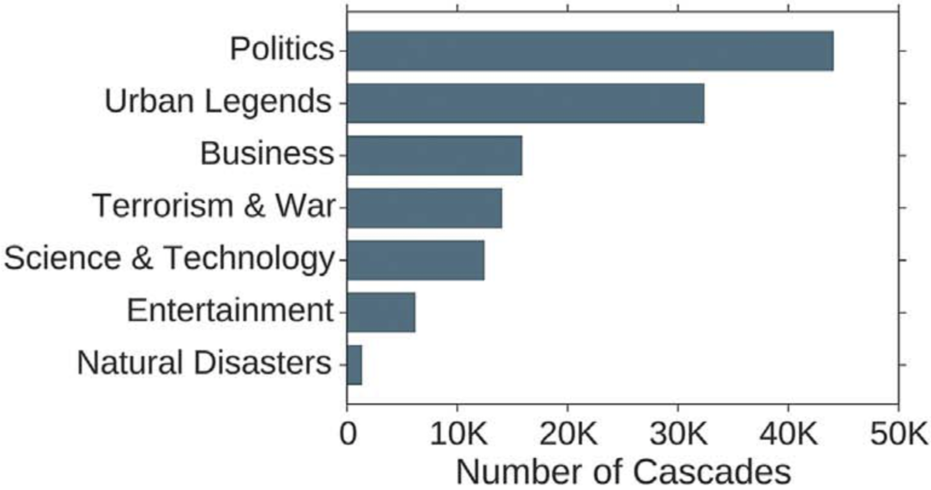
- Now researchers can vary the environment (norms) and feedback (reinforcement)



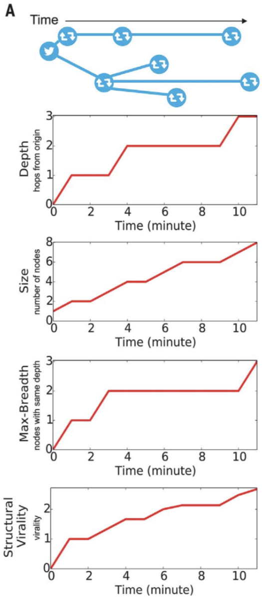
- Norms matter and people learn to choose outrage

The spread of true and false news online

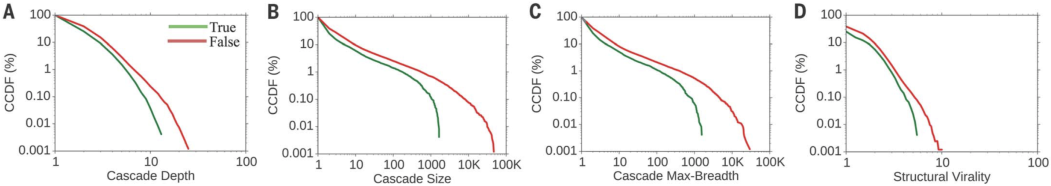
Soroush Vosoughi,[1] Deb Roy,[1] Sinan Aral[2]*



Measured as true or false based on 6 fact checking websites



A
Time
Depth
hops from origin
Time (minute)
Size
number of nodes
Time (minute)
Max-Breadth
nodes with same depth
Time (minute)
Structural Virality
virality
Time (minute)



- false rumors spread deeper, are more retweeted, spread more broadly, and are more viral than true rumors

What might explain this pattern?

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Robustness checks

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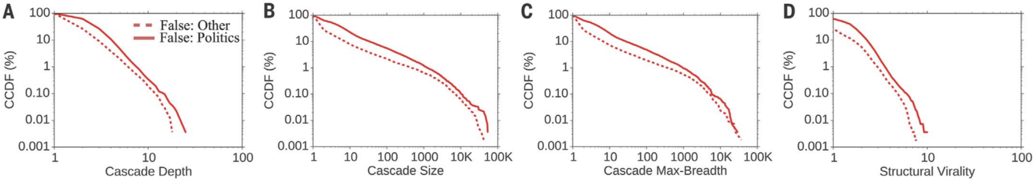
- Is this because the fact checked rumors are somehow different? No. Newly, independently checked rumors show similar pattern.
- Is this because of bots? No. If you remove bots you get similar patterns



- political rumors spreads deeper, are more retweeted, spread more broadly, and are more viral than true rumors

All of this might make you think that we are awash in false rumors about politics, but. . . . .

Fake news on Twitter during the 2016 U.S. presidential election

Nir Grinberg[1,2]*, Kenneth Joseph[3]*, Lisa Friedland[1]*, Briony Swire-Thompson[1,2], David Lazer[1,2]†

The spread of fake news on social media became a public concern in the United States after the 2016 presidential election. We examined exposure to and sharing of fake news by registered voters on Twitter and found that engagement with fake news sources was extremely concentrated. Only 1% of individuals accounted for 80% of fake news source exposures, and 0.1% accounted for nearly 80% of fake news sources shared. Individuals most likely to engage with fake news sources were conservative leaning, older, and highly engaged with political news. A cluster of fake news sources shared overlapping audiences on the extreme right, but for people across the political spectrum, most political news exposure still came from mainstream media outlets.

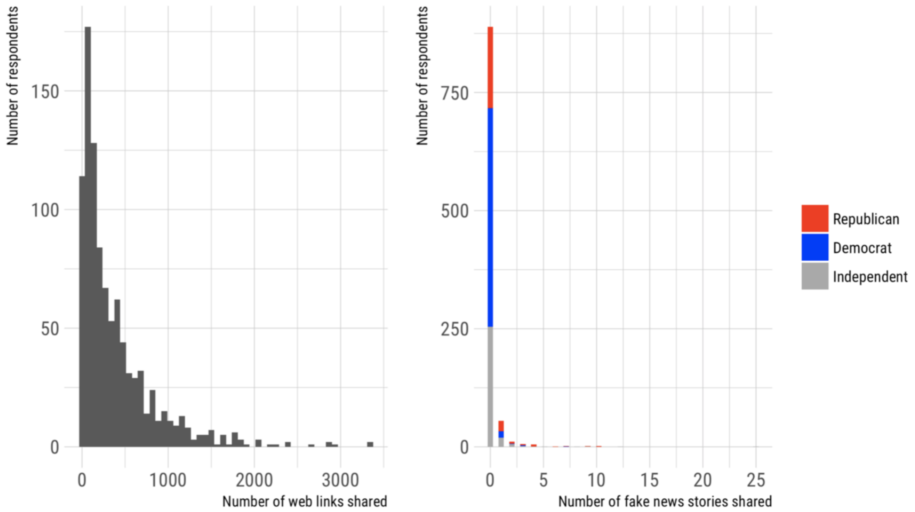
Twitter, 2016 US election, `https://dx.doi.org/10.1126/science.aau2706`

Less than you think: Prevalence and predictors of fake news dissemination on Facebook

Andrew Guess[1]*, Jonathan Nagler[2], Joshua Tucker[2]

So-called "fake news" has renewed concerns about the prevalence and effects of misinformation in political campaigns. Given the potential for widespread dissemination of this material, we examine the individual-level characteristics associated with sharing false articles during the 2016 U.S. presidential campaign. To do so, we uniquely link an original survey with respondents' sharing activity as recorded in Facebook profile data. First and foremost, we find that sharing this content was a relatively rare activity. Conservatives were more likely to share articles from fake news domains, which in 2016 were largely pro-Trump in orientation, than liberals or moderates. We also find a strong age effect, which persists after controlling for partisanship and ideology: On average, users over 65 shared nearly seven times as many articles from fake news domains as the youngest age group.

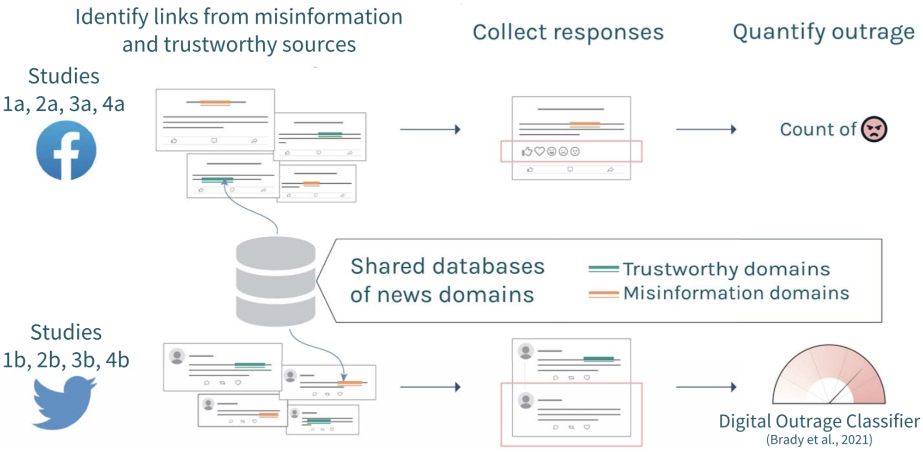
Facebook, 2016 US election, `https://dx.doi.org/10.1126/sciadv.aau4586`



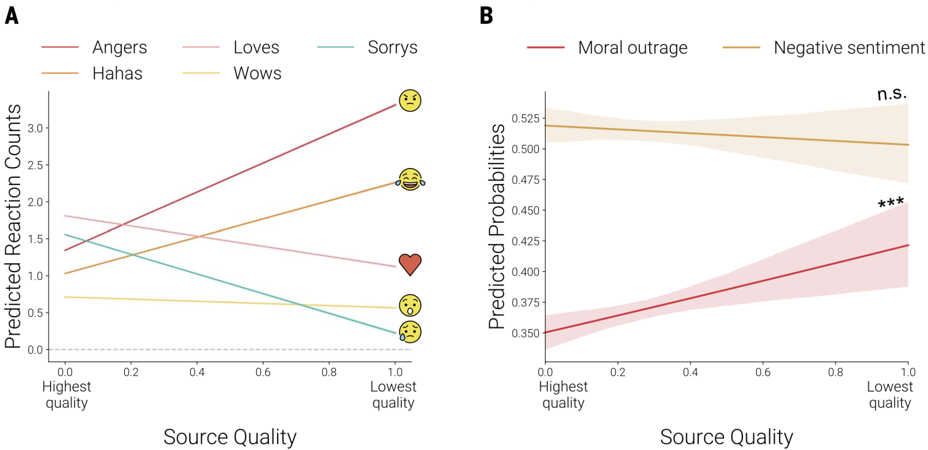
- people share links of Facebook, just not fake news

Misinformation exploits outrage to spread online

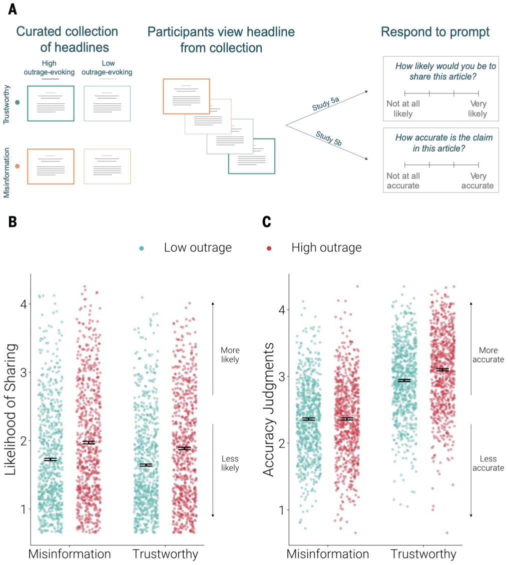
Killian L. McLoughlin[1,2]†, William J. Brady[3]*†, Aden Goolsbee[4], Ben Kaiser[5], Kate Klonick[6,7,8,9], M. J. Crockett[1,5,10]*



- important to study more than one platform



- low quality sources are more likely to evoke outrage on Facebook (A) and Twitter (B)



- Higher outrage increases sharing but does not affect accuracy discernment

Why do these studies matter? (Vosoughi et al and McLoughlin et al)

- They impact which policy you might use to intervene (eg, bot removal; focus on epistemic vs nonepistemic motives for sharing)

Stepping back:

- Social media seems to amplify moral outrage and false rumors

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- Social media seems to amplify moral outrage and false rumors
- This happens because of the choices of people and the choices of social media architects

Next class: Social media and democracy